Literal structural descriptions in fourteen retained $L_2(q)$ tables

Private recovery component

Theorem 1. *For the exact current companion tables and the accepted numerical row bijections listed below, all 348 displayed structural descriptions hold for their actual retained subgroup representatives. This asserts the indicated decompositions, not that an extension label determines a unique group.*

q	7	8	11	13	17	19
rows	15	12	16	16	22	19
q	16	23	25	27	29	31
rows	21	23	37	16	22	29
q	32	64				
rows	24	76				

The tables are attributed to Thomas Connor and Dimitri Leemans. All nineteen native pages were inspected. Their dates range from November 2012 to June 2013, but identity with every historical file available to the 2013 source article is not claimed. The retained numerical classifications and weighted incidence checks are separate acquired components. The term "lattice" in the source headings does not supply a lattice property here.

The token trees in `native_assertions.json` preserve the literal punctuation and left association: a colon denotes the displayed split extension, a centered dot the displayed nonsplit extension, and p^e an elementary abelian group. Bare integers denote cyclic groups. In particular, row 5 of $L_2(32)$ is cyclic of order 33, whereas row 4 of $L_2(27)$ is 3^3. The dihedral subscript denotes group order. No exponent is recovered by guessing how to split a flattened decimal string.

Retained inputs. The acquired complete inventories supply actual permutation generators for all subgroup representatives and the exact weighted row bijections. The robust cached-weight calculation supplies the final orders and incidences; no previous local-class multiplicity assumption is reintroduced. The fourteen top rows use the original executed $\mathrm{PSL}(2,q)$ constructors, their literal saved ambient models and the full-order subgroup in each accepted inventory. A subgroup of a finite group having the full group order equals that group. Its selected generators need not be the constructor's generators. This route retains the named-constructor identification premise; order alone is never used to identify an arbitrary group with $L_2(q)$.

The registry acquires the 29 exact small models and decomposition certificates from the accepted A5/M11 structural components. It also uses the literal retained $L_2(8)$ model for the proper occurrence in $L_2(64)$. The latter occurrence is paid by an explicit isomorphism, not by an ambient name. The generator-edge transport theorem MC.3 in the accepted marked-coordinate umbrella is acquired in full. No ambient subgroup census is run here.

Efficient literal transport. All permutations act on the right: ab sends i first to i^a and then to $(i^a)^b$. Each concrete registry model has ordered generators and a complete positive-word closure. A stored vector f is checked to map identity to identity, to be a bijection onto the closure of the actual target generators, and to satisfy $f(xa) = f(x)f(a)$ on every right-generator edge. Induction on words gives the unique homomorphism; bijectivity makes it an isomorphism. This is precisely acquired MC.3. The cost is proportional to the number of group elements times the number of generators, not to a full multiplication table. New models are indexed by their literal data, never by a supposedly canonical extension name. Factor references point only to previously proved models, so the certificate dependency order is acyclic.

Atoms and split extensions. A cyclic atom has an element of the asserted order in a literal group of that order. An elementary abelian atom has the asserted number of commuting order-p generators and their complete closure has order p^e. A dihedral atom has r, s with orders $m, 2$, $srs = r^{-1}$, and their closure has order $2m$. These conditions prove the displayed atom. The A_4, A_5, S_3 atoms and the already certified small split types use the acquired registry.

At a new split node $H = N : K$, the verifier constructs N and K from their actual generators, checks membership in H, and transports their recursively certified descriptions. Conjugation of each generator of N by each generator of H remains in N. Finiteness makes these inclusions equalities and proves $N \trianglelefteq H$. It checks $N \cap K = 1$ and $|N||K| = |H|$; hence $H = NK$ and $H \cong N \rtimes K$. A noncommuting generator pair additionally pays the displayed nontrivial action. This does not assert that all actions with the same factors are isomorphic. Every nested node follows the exact source tree.

Lemma 1. *Let $N \trianglelefteq H$ have prime index p. The quotient map onto H/N splits if and only if some element outside N has order p.*

Proof. A complement has order p, so any of its nonidentity elements gives such an element. Conversely, an outside element of order p generates a subgroup meeting N trivially and mapping onto the quotient. It is a complement. □

For each centered-dot node in this cluster, the quotient has order 2 or 3. The actual normal subgroup and its complete factor certificate are checked as above. The verifier then checks $x^p \neq 1$ for *every* element of $H \setminus N$. This proves nonsplitting by the lemma. Failure to find a complement is not used as evidence. The prime-three case is the elementary extension of the previously accepted prime-two criterion.

Complete certificate and controls. The deterministic literal verifier pays all 334 proper-subgroup occurrences, with 70 new concrete registry models in addition to the 30 acquired model inputs. There are 387 occurrence/factor transports, 69,818 right-generator edge checks and 3,488 outside-element checks for nonsplitting. Together with the fourteen exact top-model routes, these account for all 348 rows. The source join checks every original generator array and numerical row map; none of the source names filtered an ambient group search.

Candidate generation only sought witnesses inside the retained proper subgroups. It completed in 3.7 seconds under a fixed total 120-second budget. GAP normal-subgroup and isomorphism routines suggested these witnesses; their search completeness is not a premise of the literal certificate. A startup failure caused by an empty generator list was preserved before the successful pass. One successful GAP output split an integer by its standard backslash-newline continuation. The raw file is preserved and the decoder permits removal only when both adjacent characters are decimal digits. No mathematical search was repeated for that formatting issue.

The controls damage identity images, a bijective nonhomomorphism, element coverage, source-row coverage, actual generator and row-map joins, source superscripts and punctuation, quotient orders, elementary bases, dihedral relations and the factor dependency order. All must be rejected. Separate literal controls distinguish the nonsplit cyclic groups of orders 4 and 9 from split elementary abelian groups of those orders. Ordinary and optimized Python execute the same explicit checks; assertions are not used as gates.

Scope and attribution. This is a finite structural-label recovery on the fourteen specified current tables, conditional on the already acquired complete inventories and explicit named-constructor inputs. It proves neither a new classification of subgroups of $L_2(q)$ nor the identities of unexamined historical editions. It does not complete the entire subgroup atlas or add a new original-source, conjecture, formalization or measurement credit. The normal-series, semidirect-product and generator-edge arguments are standard, with their existing project interfaces retained. Root suggested the cluster/registry strategy and reuse of MC.3; the author prepared the native ledger, witnesses and literal checker.